Module 3: Logistic Regression — Completed Solutions

This solution follows the uploaded notebook and fills its incomplete sigmoid and scikit-learn training sections. The dataset is the six-example dataset supplied in the notebook.

1. Sigmoid / Logistic Function

Formula: g(z) = 1 / (1 + e^(-z)). It maps any real input to a value between 0 and 1. Use NumPy so the function works for scalars and arrays.

```
def sigmoid(z):
    """Compute the sigmoid of a scalar or NumPy array."""
    g = 1 / (1 + np.exp(-z))
    return g
```

For example, sigmoid(0) = 0.5. Large negative inputs approach 0; large positive inputs approach 1.

2. Decision Boundary

The notebook gives w0 = 1, w1 = 1 and b = -3, so h(x) = sigmoid(x0 + x1 - 3). With the 0.5 threshold, predict class 1 when x0 + x1 - 3 ≥ 0, and class 0 otherwise.

Setting the boundary expression to zero gives x0 + x1 - 3 = 0, or x1 = 3 - x0. This straight line separates the two prediction regions.

3. Train a Logistic Regression Model with scikit-learn

Complete the training section with the following code. The model is fitted to X and y, then used to predict labels and report its accuracy, coefficients, and intercept.

```
from sklearn.linear_model import LogisticRegression

# Create and train the model
model = LogisticRegression()
model.fit(X, y.ravel())

# Predict labels and calculate training accuracy
predictions = model.predict(X)
score = model.score(X, y.ravel())

# Model parameters
w0, w1 = model.coef_[0]
b = model.intercept_[0]

print("Predictions:", predictions)
print("Training accuracy:", score)
print("Coefficients (w0, w1):", w0, w1)
print("Intercept (b):", b)
```

4. Results for the supplied dataset

Using scikit-learn's default LogisticRegression settings in this environment:

```
Predictions: [0, 0, 0, 1, 1, 1]
Training accuracy: 1.0000 (100.00%)
Coefficients (w0, w1): 0.904113, 0.735875
Intercept (b): -2.333716
```

The fitted coefficients and intercept are learned from the data, so they are not expected to equal the illustrative parameters (1, 1, -3) used earlier in the notebook. Exact parameter values can vary slightly across scikit-learn versions and solver settings.

5. Full key code cells

```
# Sigmoid function
def sigmoid(z):
    return 1 / (1 + np.exp(-z))

# Example dataset from the notebook
X = np.array([
    [0.5, 1.5],
    [1, 1],
    [1.5, 0.5],
    [3, 0.5],
    [2, 2],
    [1, 2.5]
])
y = np.array([0, 0, 0, 1, 1, 1]).reshape(-1, 1)

# Train logistic regression
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
model.fit(X, y.ravel())

predictions = model.predict(X)
score = model.score(X, y.ravel())
w0, w1 = model.coef_[0]
b = model.intercept_[0]
```

6. What the output means

• Predictions are the model's predicted class (0 or 1) for each training example.

• Training accuracy is the fraction of training examples classified correctly.

• Coefficients w0 and w1 measure how each feature affects the model's log-odds.

• The intercept b is the model's bias term.

• The sigmoid output is interpreted as an estimated probability for class 1; the usual decision threshold is 0.5.